

Session-4-exercises.R

```
#####  
### Session 4 Exercises ###  
#####  
  
# Separate  
#a  
# Load the tidyverse and read in the Medicare payments dataset  
  
names <- c("DRG", "ProviderID", "Name", "Address", "City", "State", "ZIP",  
           "Region", "Discharges", "AverageCharges", "AverageTotalPayments",  
           "AverageMedicarePayments")  
types = 'ccccccinnn'  
inpatient <- read_tsv('inpatient.tsv',  
                     col_names = names, skip=1, col_types = types)  
  
#b  
# Take a Look at the diagnosis-related group (DRG) unique values  
unique(inpatient$DRG)  
  
## [1] "039 - EXTRACRANIAL PROCEDURES W/O CC/MCC"  
## [2] "057 - DEGENERATIVE NERVOUS SYSTEM DISORDERS W/O MCC"  
## [3] "069 - TRANSIENT ISCHEMIA"  
## [4] "064 - INTRACRANIAL HEMORRHAGE OR CEREBRAL INFARCTION W MCC"  
## [5] "065 - INTRACRANIAL HEMORRHAGE OR CEREBRAL INFARCTION W CC"  
## [6] "066 - INTRACRANIAL HEMORRHAGE OR CEREBRAL INFARCTION W/O CC/MCC"  
## [7] "074 - CRANIAL & PERIPHERAL NERVE DISORDERS W/O MCC"  
## [8] "101 - SEIZURES W/O MCC"  
## [9] "149 - DYSEQUILIBRIUM"  
## [10] "176 - PULMONARY EMBOLISM W/O MCC"  
## [11] "177 - RESPIRATORY INFECTIONS & INFLAMMATIONS W MCC"  
## [12] "189 - PULMONARY EDEMA & RESPIRATORY FAILURE"  
## [13] "178 - RESPIRATORY INFECTIONS & INFLAMMATIONS W CC"  
## [14] "190 - CHRONIC OBSTRUCTIVE PULMONARY DISEASE W MCC"  
## [15] "191 - CHRONIC OBSTRUCTIVE PULMONARY DISEASE W CC"  
## [16] "193 - SIMPLE PNEUMONIA & PLEURISY W MCC"  
## [17] "194 - SIMPLE PNEUMONIA & PLEURISY W CC"  
## [18] "192 - CHRONIC OBSTRUCTIVE PULMONARY DISEASE W/O CC/MCC"  
## [19] "195 - SIMPLE PNEUMONIA & PLEURISY W/O CC/MCC"  
## [20] "202 - BRONCHITIS & ASTHMA W CC/MCC"  
## [21] "203 - BRONCHITIS & ASTHMA W/O CC/MCC"  
## [22] "207 - RESPIRATORY SYSTEM DIAGNOSIS W VENTILATOR SUPPORT 96+ HOURS"  
## [23] "238 - MAJOR CARDIOVASC PROCEDURES W/O MCC"  
## [24] "252 - OTHER VASCULAR PROCEDURES W MCC"  
## [25] "253 - OTHER VASCULAR PROCEDURES W CC"  
## [26] "208 - RESPIRATORY SYSTEM DIAGNOSIS W VENTILATOR SUPPORT <96 HOURS"  
## [27] "254 - OTHER VASCULAR PROCEDURES W/O CC/MCC"  
## [28] "291 - HEART FAILURE & SHOCK W MCC"  
## [29] "243 - PERMANENT CARDIAC PACEMAKER IMPLANT W CC"  
## [30] "244 - PERMANENT CARDIAC PACEMAKER IMPLANT W/O CC/MCC"
```

[31] "246 - PERC CARDIOVASC PROC W DRUG-ELUTING STENT W MCC OR 4+ VESSELS/STENTS"
[32] "247 - PERC CARDIOVASC PROC W DRUG-ELUTING STENT W/O MCC"
[33] "249 - PERC CARDIOVASC PROC W NON-DRUG-ELUTING STENT W/O MCC"
[34] "251 - PERC CARDIOVASC PROC W/O CORONARY ARTERY STENT W/O MCC"
[35] "280 - ACUTE MYOCARDIAL INFARCTION, DISCHARGED ALIVE W MCC"
[36] "292 - HEART FAILURE & SHOCK W CC"
[37] "281 - ACUTE MYOCARDIAL INFARCTION, DISCHARGED ALIVE W CC"
[38] "282 - ACUTE MYOCARDIAL INFARCTION, DISCHARGED ALIVE W/O CC/MCC"
[39] "286 - CIRCULATORY DISORDERS EXCEPT AMI, W CARD CATH W MCC"
[40] "287 - CIRCULATORY DISORDERS EXCEPT AMI, W CARD CATH W/O MCC"
[41] "293 - HEART FAILURE & SHOCK W/O CC/MCC"
[42] "300 - PERIPHERAL VASCULAR DISORDERS W CC"
[43] "301 - PERIPHERAL VASCULAR DISORDERS W/O CC/MCC"
[44] "303 - ATHEROSCLEROSIS W/O MCC"
[45] "305 - HYPERTENSION W/O MCC"
[46] "308 - CARDIAC ARRHYTHMIA & CONDUCTION DISORDERS W MCC"
[47] "312 - SYNCOPE & COLLAPSE"
[48] "309 - CARDIAC ARRHYTHMIA & CONDUCTION DISORDERS W CC"
[49] "310 - CARDIAC ARRHYTHMIA & CONDUCTION DISORDERS W/O CC/MCC"
[50] "313 - CHEST PAIN"
[51] "314 - OTHER CIRCULATORY SYSTEM DIAGNOSES W MCC"
[52] "315 - OTHER CIRCULATORY SYSTEM DIAGNOSES W CC"
[53] "330 - MAJOR SMALL & LARGE BOWEL PROCEDURES W CC"
[54] "377 - G.I. HEMORRHAGE W MCC"
[55] "329 - MAJOR SMALL & LARGE BOWEL PROCEDURES W MCC"
[56] "372 - MAJOR GASTROINTESTINAL DISORDERS & PERITONEAL INFECTIONS W CC"
[57] "378 - G.I. HEMORRHAGE W CC"
[58] "379 - G.I. HEMORRHAGE W/O CC/MCC"
[59] "389 - G.I. OBSTRUCTION W CC"
[60] "390 - G.I. OBSTRUCTION W/O CC/MCC"
[61] "391 - ESOPHAGITIS, GASTROENT & MISC DIGEST DISORDERS W MCC"
[62] "394 - OTHER DIGESTIVE SYSTEM DIAGNOSES W CC"
[63] "439 - DISORDERS OF PANCREAS EXCEPT MALIGNANCY W CC"
[64] "392 - ESOPHAGITIS, GASTROENT & MISC DIGEST DISORDERS W/O MCC"
[65] "460 - SPINAL FUSION EXCEPT CERVICAL W/O MCC"
[66] "473 - CERVICAL SPINAL FUSION W/O CC/MCC"
[67] "418 - LAPAROSCOPIC CHOLECYSTECTOMY W/O C.D.E. W CC"
[68] "419 - LAPAROSCOPIC CHOLECYSTECTOMY W/O C.D.E. W/O CC/MCC"
[69] "469 - MAJOR JOINT REPLACEMENT OR REATTACHMENT OF LOWER EXTREMITY W MCC"
[70] "470 - MAJOR JOINT REPLACEMENT OR REATTACHMENT OF LOWER EXTREMITY W/O MCC"
[71] "480 - HIP & FEMUR PROCEDURES EXCEPT MAJOR JOINT W MCC"
[72] "481 - HIP & FEMUR PROCEDURES EXCEPT MAJOR JOINT W CC"
[73] "536 - FRACTURES OF HIP & PELVIS W/O MCC"
[74] "482 - HIP & FEMUR PROCEDURES EXCEPT MAJOR JOINT W/O CC/MCC"
[75] "552 - MEDICAL BACK PROBLEMS W/O MCC"
[76] "491 - BACK & NECK PROC EXC SPINAL FUSION W/O CC/MCC"
[77] "563 - FX, SPRN, STRN & DISL EXCEPT FEMUR, HIP, PELVIS & THIGH W/O MCC"
[78] "602 - CELLULITIS W MCC"
[79] "603 - CELLULITIS W/O MCC"
[80] "638 - DIABETES W CC"
[81] "640 - MISC DISORDERS OF NUTRITION,METABOLISM,FLUIDS/ELECTROLYTES W MCC"
[82] "682 - RENAL FAILURE W MCC"
[83] "641 - MISC DISORDERS OF NUTRITION,METABOLISM,FLUIDS/ELECTROLYTES W/O MCC"
[84] "683 - RENAL FAILURE W CC"
[85] "684 - RENAL FAILURE W/O CC/MCC"

```
## [86] "689 - KIDNEY & URINARY TRACT INFECTIONS W MCC"
## [87] "690 - KIDNEY & URINARY TRACT INFECTIONS W/O MCC"
## [88] "811 - RED BLOOD CELL DISORDERS W MCC"
## [89] "698 - OTHER KIDNEY & URINARY TRACT DIAGNOSES W MCC"
## [90] "699 - OTHER KIDNEY & URINARY TRACT DIAGNOSES W CC"
## [91] "812 - RED BLOOD CELL DISORDERS W/O MCC"
## [92] "853 - INFECTIOUS & PARASITIC DISEASES W O.R. PROCEDURE W MCC"
## [93] "885 - PSYCHOSES"
## [94] "870 - SEPTICEMIA OR SEVERE SEPSIS W MV 96+ HOURS"
## [95] "871 - SEPTICEMIA OR SEVERE SEPSIS W/O MV 96+ HOURS W MCC"
## [96] "872 - SEPTICEMIA OR SEVERE SEPSIS W/O MV 96+ HOURS W/O MCC"
## [97] "897 - ALCOHOL/DRUG ABUSE OR DEPENDENCE W/O REHABILITATION THERAPY W/O MCC"
## [98] "948 - SIGNS & SYMPTOMS W/O MCC"
## [99] "917 - POISONING & TOXIC EFFECTS OF DRUGS W MCC"
## [100] "918 - POISONING & TOXIC EFFECTS OF DRUGS W/O MCC"
```

#c

Let's try separating this on the hyphen

```
inpatient_separate <- separate(inpatient,DRG,c('DRGcode','DRGdescription'),'-' )
```

```
## Warning: Expected 2 pieces. Additional pieces discarded in 3280 rows [45894,
45895,
## 45896, 45897, 45898, 45899, 45900, 45901, 45902, 45903, 45904, 45905, 45906,
## 45907, 45908, 45909, 45910, 45911, 45912, 45913, ...].
```

#d

Take a Look at the error message and try to find out what is going on.

Let's Look at row 45894

```
inpatient$DRG[45894]
```

```
## [1] "246 - PERC CARDIOVASC PROC W DRUG-ELUTING STENT W MCC OR 4+ VESSELS/STENTS"
```

#Maybe

```
inpatient_separate <- separate(inpatient,DRG,c('DRGcode','DRGdescription'),'-' )
```

#e

Let's separate with character position instead

```
inpatient_separate <- separate(inpatient,DRG,c('DRGcode','DRGdescription'),4)
```

#f

Take a Look at the data now

```
glimpse(inpatient_separate)
```

```
## Rows: 163,065
## Columns: 13
## $ DRGcode      <chr> "039 ", "039 ", "039 ", "039 ", "039 ", "039 "...
## $ DRGdescription <chr> "- EXTRACRANIAL PROCEDURES W/O CC/MCC", "- EXT...
## $ ProviderID    <chr> "10001", "10005", "10006", "10011", "10016", "...
## $ Name          <chr> "SOUTHEAST ALABAMA MEDICAL CENTER", "MARSHALL ...
## $ Address       <chr> "1108 ROSS CLARK CIRCLE", "2505 U S HIGHWAY 43...
## $ City          <chr> "DOTHAN", "BOAZ", "FLORENCE", "BIRMINGHAM", "A...
## $ State         <chr> "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL"...
```

```

## $ ZIP                <chr> "36301", "35957", "35631", "35235", "35007", "...
## $ Region             <chr> "AL - Dothan", "AL - Birmingham", "AL - Birmin...
## $ Discharges         <int> 91, 14, 24, 25, 18, 67, 51, 32, 135, 34, 14, 4...
## $ AverageCharges     <dbl> 32963.07, 15131.85, 37560.37, 13998.28, 31633.0...
## $ AverageTotalPayments <dbl> 5777.24, 5787.57, 5434.95, 5417.56, 5658.33, 6...
## $ AverageMedicarePayments <dbl> 4763.73, 4976.71, 4453.79, 4129.16, 4851.44, 5...

inpatient_separate <- separate(inpatient, DRG, c('DRGcode', 'DRGdescription'), 3)
inpatient_separate <-
separate(inpatient_separate, DRGdescription, c('waste', 'DRGdescription'), 3)

inpatient_separate <- select(inpatient_separate, -waste)

# The same with the new function:
#inpatient_separate <- separate_wider_delim(inpatient,
#                                           cols=DRG,
#                                           names=c('DRGcode', 'DRGdescription'),
#                                           delim='-')
#It does not work, we have to tell what to do with problematic values

inpatient_separate <- separate_wider_delim(inpatient,
                                           cols=DRG,
                                           names=c('DRGcode', 'DRGdescription'),
                                           delim='-',
                                           too_many = "merge")
#It makes the job. Not perfectly, " - " has not been trimmed.

#Position works quite badly:
inpatient_separate <- separate_wider_position(inpatient,
                                           cols=DRG,
                                           widths=c(DRGcode = 3, 3,
DRGdescription=200),
                                           too_few="debug")

## Warning: Debug mode activated: adding variables `DRG_ok`, `DRG_width`, and
## `DRG_remainder`.

#It gives the good results, but we have to remove the added columns:
inpatient_separate <- inpatient_separate %>% select(-c('DRG', 'DRG_ok',
'DRG_width', 'DRG_remainder'))

```

```

#Unite
# Load the tidyverse and the food inspections dataset
library(tidyverse)

names <- c("ID", "DBAName", "AKAName", "License", "FacilityType", "Risk",
"Address", "City", "State", "ZIP", "InspectionDate", "InspectionType",
"Results", "Violations", "Latitude", "Longitude", "Location")

inspections <- read_csv('inspections.csv',
                        col_names=names, skip=1)

## Rows: 145606 Columns: 17
## — Column specification —————
## Delimiter: ","
## chr (12): DBAName, AKAName, FacilityType, Risk, Address, City, State, Inspec...
## dbl (5): ID, License, ZIP, Latitude, Longitude
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

glimpse(inspections)

## Rows: 145,606
## Columns: 17
## $ ID <dbl> 1995829, 1995817, 1995822, 1995814, 1995811, 1995752, 1...
## $ DBAName <chr> "SUBWAY", "VENEZUELAN BOWL GRILL LLC", "SEVEN TEN", "CH...
## $ AKAName <chr> "SUBWAY", "VENEZUELAN BOWL GRILL", "SEVEN TEN", "CHISME...
## $ License <dbl> 1679112, 2424110, 1172093, 1334960, 2334190, 2517730, 2...
## $ FacilityType <chr> "Restaurant", "Restaurant", "Restaurant", "Restaurant",...
## $ Risk <chr> "Risk 1 (High)", "Risk 1 (High)", "Risk 1 (High)", "Ris...
## $ Address <chr> "8711 S ASHLAND AVE", "2436 N LINCOLN AVE", "1055 E 55T...
## $ City <chr> "CHICAGO", "CHICAGO", "CHICAGO", "CHICAGO", "CHICAGO", ...
## $ State <chr> "IL", "IL", "IL", "IL", "IL", "IL", "IL", "IL", "IL", "IL", ...
## $ ZIP <dbl> 60620, 60614, 60615, 60629, 60615, 60621, 60630, 60614,...
## $ InspectionDate <chr> "03/17/2017", "03/17/2017", "03/17/2017", "03/17/2017",...
## $ InspectionType <chr> "Canvass Re-Inspection", "Canvass Re-Inspection", "Canv...
## $ Results <chr> "Pass", "Pass", "Pass", "Pass", "Pass", "Pass", "Pass", "Out of...
## $ Violations <chr> "19. OUTSIDE GARBAGE WASTE GREASE AND STORAGE AREA; CLE...
## $ Latitude <dbl> 41.73541, 41.92619, 41.79492, 41.78429, 41.79517, 41.77...
## $ Longitude <dbl> -87.66291, -87.65002, -87.59985, -87.72277, -87.59710, ...
## $ Location <chr> "(41.735408702583165, -87.66290968424336)", "(41.926190..."

# Create a new column called Regions that combines City and State
regional_inspections <- unite(inspections, Region, City, State, sep=", ")

# Let's Look at the data
glimpse(regional_inspections)

## Rows: 145,606
## Columns: 16
## $ ID <dbl> 1995829, 1995817, 1995822, 1995814, 1995811, 1995752, 1...
## $ DBAName <chr> "SUBWAY", "VENEZUELAN BOWL GRILL LLC", "SEVEN TEN", "CH...
## $ AKAName <chr> "SUBWAY", "VENEZUELAN BOWL GRILL", "SEVEN TEN", "CHISME...

```

```
## $ License      <dbl> 1679112, 2424110, 1172093, 1334960, 2334190, 2517730, 2...
## $ FacilityType <chr> "Restaurant", "Restaurant", "Restaurant", "Restaurant",...
## $ Risk         <chr> "Risk 1 (High)", "Risk 1 (High)", "Risk 1 (High)", "Ris...
## $ Address      <chr> "8711 S ASHLAND AVE", "2436 N LINCOLN AVE", "1055 E 55T...
## $ Region       <chr> "CHICAGO, IL", "CHICAGO, IL", "CHICAGO, IL", "CHICAGO, ...
## $ ZIP          <dbl> 60620, 60614, 60615, 60629, 60615, 60621, 60630, 60614,...
## $ InspectionDate <chr> "03/17/2017", "03/17/2017", "03/17/2017", "03/17/2017",...
## $ InspectionType <chr> "Canvass Re-Inspection", "Canvass Re-Inspection", "Canv...
## $ Results      <chr> "Pass", "Pass", "Pass", "Pass", "Pass", "Pass", "Out of...
## $ Violations   <chr> "19. OUTSIDE GARBAGE WASTE GREASE AND STORAGE AREA; CLE...
## $ Latitude     <dbl> 41.73541, 41.92619, 41.79492, 41.78429, 41.79517, 41.77...
## $ Longitude    <dbl> -87.66291, -87.65002, -87.59985, -87.72277, -87.59710, ...
## $ Location     <chr> "(41.735408702583165, -87.66290968424336)", "(41.926190...
```

Whoops. I didn't want to DELETE the City and State columns. Let's try again.

```
regional_inspections <- unite(inspections, Region, City, State, sep=" ",
remove=FALSE)
glimpse(regional_inspections)
```

```
## Rows: 145,606
## Columns: 18
## $ ID          <dbl> 1995829, 1995817, 1995822, 1995814, 1995811, 1995752, 1...
## $ DBAName     <chr> "SUBWAY", "VENEZUELAN BOWL GRILL LLC", "SEVEN TEN", "CH...
## $ AKAName     <chr> "SUBWAY", "VENEZUELAN BOWL GRILL", "SEVEN TEN", "CHISME...
## $ License     <dbl> 1679112, 2424110, 1172093, 1334960, 2334190, 2517730, 2...
## $ FacilityType <chr> "Restaurant", "Restaurant", "Restaurant", "Restaurant",...
## $ Risk        <chr> "Risk 1 (High)", "Risk 1 (High)", "Risk 1 (High)", "Ris...
## $ Address     <chr> "8711 S ASHLAND AVE", "2436 N LINCOLN AVE", "1055 E 55T...
## $ Region      <chr> "CHICAGO, IL", "CHICAGO, IL", "CHICAGO, IL", "CHICAGO, ...
## $ City        <chr> "CHICAGO", "CHICAGO", "CHICAGO", "CHICAGO", "CHICAGO", ...
## $ State       <chr> "IL", "IL", "IL", "IL", "IL", "IL", "IL", "IL", "IL", "..."
## $ ZIP         <dbl> 60620, 60614, 60615, 60629, 60615, 60621, 60630, 60614,...
## $ InspectionDate <chr> "03/17/2017", "03/17/2017", "03/17/2017", "03/17/2017",...
## $ InspectionType <chr> "Canvass Re-Inspection", "Canvass Re-Inspection", "Canv...
## $ Results     <chr> "Pass", "Pass", "Pass", "Pass", "Pass", "Pass", "Out of...
## $ Violations  <chr> "19. OUTSIDE GARBAGE WASTE GREASE AND STORAGE AREA; CLE...
## $ Latitude    <dbl> 41.73541, 41.92619, 41.79492, 41.78429, 41.79517, 41.77...
## $ Longitude   <dbl> -87.66291, -87.65002, -87.59985, -87.72277, -87.59710, ...
## $ Location    <chr> "(41.735408702583165, -87.66290968424336)", "(41.926190...
```

And take a Look at the unique regions

```
unique(regional_inspections$Region)
```

```
## [1] "CHICAGO, IL"      "NA, IL"
## [3] "ELK GROVE VILLAGE, IL" "SCHAUMBURG, IL"
## [5] "Chicago, IL"      "CALUMET CITY, IL"
## [7] "SKOKIE, IL"       "chicago, IL"
## [9] "EVANSTON, IL"     "CCHICAGO, IL"
## [11] "NILES NILES, IL"  "Chicago, IL"
## [13] "ELMHURST, IL"    "NA, NA"
## [15] "SCHILLER PARK, IL" "OAK PARK, IL"
## [17] "BRIDEVIEW, IL"   "CHESTNUT STREET, IL"
## [19] "BEDFORD PARK, IL" "BANNOCKBURNDEERFIELD, IL"
## [21] "CHCICAGO, IL"    "BLOOMINGDALE, IL"
```

```
## [23] "CICERO, IL" "Norridge, IL"
## [25] "CHICAGO, NA" "CHARLES A HAYES, IL"
## [27] "CHICAGOCHICAGO, IL" "MAYWOOD, IL"
## [29] "CHCHICAGO, IL" "CHICAGOI, IL"
## [31] "SUMMIT, IL" "OOLYMPIA FIELDS, IL"
## [33] "WESTMONT, IL" "CHICAGO HEIGHTS, IL"
## [35] "JUSTICE, IL" "TINLEY PARK, IL"
## [37] "LOMBARD, IL" "EAST HAZEL CREST, IL"
## [39] "COUNTRY CLUB HILLS, IL" "STREAMWOOD, IL"
## [41] "BOLINGBROOK, IL" "INACTIVE, IL"
## [43] "BERWYN, IL" "BURNHAM, IL"
## [45] "DES PLAINES, IL" "LAKE ZURICH, IL"
## [47] "OLYMPIA FIELDS, IL" "alsip, IL"
## [49] "OAK LAWN, IL" "BLUE ISLAND, IL"
## [51] "GLENCOE, IL" "FRANKFORT, IL"
## [53] "NAPERVILLE, IL" "BROADVIEW, IL"
## [55] "WORTH, IL" "Maywood, IL"
## [57] "ALSIP, IL" "EVERGREEN PARK, IL"
```

#Exercise 4.1

```
library(tidyverse)
library(readxl)
```

```
rainfall <- read_excel("messy-data.xlsx", skip = 1)
```

```
rainfall <- rainfall %>% separate('Month, period', c("Month", "Period"), ',')
```

```
rainfall <- rainfall %>% separate('Lake Victoria', c("Lake_Victoria",
"mess"), 'm')
```

```
## Warning: Expected 2 pieces. Additional pieces discarded in 2 rows [1, 2].
```

```
## Warning: Expected 2 pieces. Missing pieces filled with `NA` in 10 rows [3, 4, 5,
6, 7, 8, 9, 10, 11, 12].
```

```
rainfall <- rainfall %>% separate(Simiyu, c("Simiyu", "mess"), 'm')
```

```
## Warning: Expected 2 pieces. Additional pieces discarded in 1 rows [2].
```

```
## Warning: Expected 2 pieces. Missing pieces filled with `NA` in 11 rows [1, 3, 4,
5, 6, 7, 8, 9, 10, 11, 12].
```

```
rainfall <- rainfall %>% select(-mess)
```

Any solution using string manipulations and Looking for patterns are OK.

```
rainfall$Lake_Victoria <- as.numeric(rainfall$Lake_Victoria)
```

```
rainfall$Simiyu <- as.numeric(rainfall$Simiyu)
```

mutate can also work

```
rainfall <- rainfall %>%
```

```
  pivot_longer(cols=c(Lake_Victoria, Simiyu), names_to="Place",
values_to="rainfall")
```

The period is useless, it is not changing, it is not a variable.

```
rainfall_2001_2019 <- rainfall %>% select(-Period)
```

#Exercise 4.2

#a

```
coffee <- read_csv("coffeeshop.csv")
```

```
## New names:
```

```
## Rows: 17 Columns: 243
```

```
## — Column specification
```

```
## _____ Delimiter: "," chr
```

```
## (243): Beverage_category, Coffee...2, Coffee...3, Coffee...4, Coffee...5...
```

```
## i Use `spec()` to retrieve the full column specification for this data. i
```

```
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
glimpse(coffee)
```

#b

Data is transposed (observations are in columns, variables in rows).

#c

```
coffee <- coffee %>%
```

```
  pivot_longer(!Beverage_category, names_to = "Beverage_type",  
               values_to = "value") %>%
```

```
  pivot_wider(names_from = Beverage_category,  
              values_from = value)
```

#d

```
#coffee <- coffee %>%
```

```
# rename(Beverage_category=Beverage_type)
```

```
coffee <- coffee %>%
```

```
  separate_wider_delim(Beverage_type, "...", names=c("Beverage_category",  
"mess"))
```

```
coffee <- coffee%>% select(-mess)
```

#Alternatively, you can use gsub and rename

#e

```
unique(coffee$`Total Fat (g)`)
```

```
## [1] "0.1" "3.5" "2.5" "0.2" "6" "4.5" "0.3" "7" "5" "0.4" "9" "1.5"
```

```
## [13] "4" "2" "8" "3" "11" "0" "1" "10" "15" "13" "0.5" "3 2"
```

A value of 3 2 instead of 3.2

```
unique(coffee$`Caffeine (mg)`)
```

```
## [1] "175" "260" "330" "410" "75" "150" "85" "95"
```

```
## [9] "180" "225" "300" "10" "20" "25" "30" "0"
```

```
## [17] "Varies" "50" "70" "120" "55" "80" "110" "varies"
```

```
## [25] "165" "235" "90" NA "125" "170" "15" "130"
```

```
## [33] "140" "100" "145" "65" "105"
```

Values like varies and Varies are not numbers.